Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: George teVelde Dairy Farm

Physical address of dairy:

27815 E Dodds RD	Escalon	San Joaquin	95320
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 07/10/2000

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X207-X130-X030-XXXX

B. OPERATORS

teVelde, George Andrew

Operator name: teVelde, George Andrew

Telephone no.: (123) 000-0000

		Landline	Cellular
27815 E Dodds RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

teVelde, George Andrew

Legal owner name: teVelde, George Andrew

Telephone no.: (123) 000-0000

		Landline	Cellular
27815 E Dodds RD	Escalon	CA	95320
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	884	912	500	580
Number under roof	2,520	342	0	0	0	0
Maximum number	2,520	342	884	912	500	580
Average number	2,520	342	884	912	500	580
Avg live weight (lbs)	1,400	1,500	100	500		

Predominant milk cow breed: Holstein

Average milk production: 97 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 96,149.21 *tons per reporting period*

Total nitrogen from manure: 1,237,605.91 *lbs per reporting period*

After ammonia losses (30% loss applied): 866,324.14 *lbs per reporting period*

Total phosphorus from manure: 203,262.41 *lbs per reporting period*

Total potassium from manure: 527,305.53 *lbs per reporting period*

Total salt from manure: 1,265,184.90 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 106,965,600 *gallons*

Total nitrogen generated: 215,914.58 *lbs*

Total phosphorus generated: 44,438.63 *lbs*

Total potassium generated: 285,158.66 *lbs*

Total salt generated: 1,441,671.52 *lbs*

	106,965,600	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	106,965,600	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #10	Ground water
Ag Well #2 River Road	Ground water
Ag Well #3 River Road	Ground water
Ag Well #4 River Road	Ground water
Ag Well/ Deep Well Dairy	Ground water

Source Description	Type
Domestic Well	Ground water
Reservoir SSJID	Surface water
Stanislaus River	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
12/31/2023	Corral solids	1,994.00 *ton*	Dry-weight	11.3		5,460.00	4,300.00	19,400.00		34.30

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	19,313.96	15,210.63	68,624.71	1,213,313.11
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	19,313.96	15,210.63	68,624.71	1,213,313.11

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
#1	80	80	2	process wastewater	X245-X026-X009-XXXX X245-X026-X011-XXXX
#10	134	134	0	none	X245-X026-X013-XXXX
#11	29	29	2	process wastewater	X245-X026-X013-XXXX
#12	15	15	2	process wastewater	X245-X026-X013-XXXX
#2	61	61	2	process wastewater	X245-X026-X009-XXXX X245-X026-X011-XXXX
#3	83	83	2	process wastewater	X245-X026-X013-XXXX
#4	33	33	2	process wastewater	X245-X026-X013-XXXX
#5	52	52	2	process wastewater	X245-X026-X013-XXXX
#6	21	21	2	process wastewater	X245-X026-X012-XXXX X245-X026-X013-XXXX
#7	62	62	2	process wastewater	X245-X026-X012-XXXX X245-X026-X013-XXXX
#8	29	29	2	process wastewater	X207-X013-X004-XXXX
#9	24	24	2	process wastewater	X245-X026-X013-XXXX
#RR-1	38	38	0	none	X245-X026-X009-XXXX X245-X026-X011-XXXX
#RR-10	48	48	0	none	X245-X026-X013-XXXX
#RR-11	15	15	0	none	X245-X026-X013-XXXX
#RR-12	14	14	0	none	X245-X026-X013-XXXX
#RR-13	5	5	0	none	X245-X026-X013-XXXX
#RR-14	4	4	0	none	X245-X026-X013-XXXX
#RR-2	34	34	0	none	X245-X026-X009-XXXX X245-X026-X011-XXXX
#RR-3	83	83	0	none	X245-X026-X013-XXXX
#RR-5	48	48	0	none	X245-X026-X013-XXXX
#RR-6	77	77	0	none	X245-X026-X012-XXXX X245-X026-X013-XXXX
#RR-7	38	38	0	none	X245-X026-X012-XXXX X245-X026-X013-XXXX
#RR-9	82	82	0	none	X245-X026-X013-XXXX
Comet 1	36	36	0	none	X207-X013-X004-XXXX
Comet 2	134	134	0	none	X207-X013-X008-XXXX

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
G1	33	33	0	none	X207-X130-X030-XXXX
G2	54	54	0	none	X207-X130-X030-XXXX
G3	24	24	0	none	X207-X130-X030-XXXX
G4	29	29	0	none	X207-X130-X030-XXXX
G5	29	29	0	none	X207-X130-X030-XXXX
L4	42	42	0	none	X207-X130-X030-XXXX
L5	42	42	0	none	X207-X130-X030-XXXX
L6	24	24	0	none	X207-X130-X030-XXXX
L7	61	61	0	none	X207-X130-X030-XXXX
South Pivot	139	139	0	none	X207-X130-X030-XXXX
Totals for areas that were used for application	489	489	22		
Totals for areas that were not used for application	1,267	1,267	0		
Land application area totals	1,756	1,756	22		

B. CROPS AND HARVESTS

#1

Field name: #1

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 80 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	928.00 *ton*	Dry-weight		80.1	33,200.00	2,560.00	22,200.00		18.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	153.28	11.82	102.49	844.87

#1

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 80 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	1,880.00 *ton*	Dry-weight		65.5	22,200.00	1,610.00	11,600.00		6.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	359.97	26.11	188.09	1,047.49

#11

Field name: #11

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	336.40 *ton*	Dry-weight		67.7	12,100.00	1,210.00	9,270.00		14.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	90.67	9.07	69.47	1,064.09

06/14/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/14/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	681.50 *ton*	Dry-weight		62.2	18,300.00	1,800.00	8,020.00		6.54

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	325.12	31.98	142.48	1,161.90

#12

Field name: #12

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	174.00 *ton*	Dry-weight		80.3	27,400.00	2,660.00	28,600.00		16.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	125.23	12.16	130.71	758.69

06/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/16/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	352.50 *ton*	Dry-weight		69.1	15,600.00	2,270.00	13,100.00		7.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	226.56	32.97	190.25	1,096.49

#2

Field name: #2

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 61 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	707.60 *ton*	Dry-weight		73.4	25,200.00	1,720.00	19,000.00		19.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	155.51	10.61	117.25	1,197.21

#2

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,433.50 *ton*	Dry-weight		65.7	13,800.00	2,510.00	10,500.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	222.47	40.46	169.27	967.26

#3

Field name: #3

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 83 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	962.80 *ton*	Dry-weight		62.9	13,400.00	1,390.00	14,200.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	115.34	11.96	122.22	1,101.72

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 83 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	1,950.50 *ton*	Dry-weight		69.3	12,300.00	1,720.00	8,260.00		5.11

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	177.48	24.82	119.18	737.32

#4

Field name: #4

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	382.80 *ton*	Dry-weight		81.1	43,900.00	2,260.00	19,500.00		19.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	192.49	9.91	85.50	850.65

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	775.50 *ton*	Dry-weight		66.3	14,200.00	2,150.00	8,600.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	224.91	34.05	136.22	760.27

#5

Field name: #5

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	382.80 *ton*	Dry-weight		68.8	8,810.00	942.00	10,300.00		13.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	63.77	6.82	74.56	977.18

#5

06/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 52 Plant date: 06/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,222.00 *ton*	Dry-weight		64.7	13,200.00	2,930.00	10,900.00		8.74

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	219.00	48.61	180.84	1,450.05

#6

Field name: #6

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	243.60 *ton*	Dry-weight		68.8	26,200.00	2,030.00	21,700.00		16.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	189.65	14.69	157.07	1,216.05

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	493.50 *ton*	Dry-weight		67.2	17,400.00	1,490.00	6,180.00		5.46

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	268.24	22.97	95.27	841.71

#7

Field name: #7

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 62 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	719.20 *ton*	Dry-weight		68.9	10,600.00	1,230.00	12,900.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	76.48	8.87	93.08	981.27

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/21/2023	1,457.00 *ton*	Dry-weight		69.1	14,600.00	2,370.00	13,000.00		7.25

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	212.04	34.42	188.80	1,052.92

#8

Field name: #8

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 29 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	336.40 *ton*	Dry-weight		77.7	32,900.00	3,090.00	34,800.00		17.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	170.21	15.99	180.04	879.51

#8

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/22/2023	681.50 *ton*	Dry-weight		62.9	11,500.00	1,740.00	8,250.00		6.66

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	200.53	30.34	143.86	1,161.30

#9

Field name: #9

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 24 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	278.40 *ton*	Dry-weight		79.0	34,100.00	3,080.00	31,000.00		16.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.60	166.14	15.01	151.03	779.52

06/13/2023: Corn, silage

Crop: Corn, silage Acres planted: 24 Plant date: 06/13/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/20/2023	564.00 *ton*	Dry-weight		68.9	16,400.00	1,950.00	9,820.00		5.04

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.50	239.72	28.50	143.54	736.70

NUTRIENT BUDGET

A. LAND APPLICATIONS

#1 - 10/10/2022: Oats, silage-soft dough

Field name: #1

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.87	3.75	59.68	350.49	960,000.00 *gal*
Application event totals		46.87	3.75	59.68	350.49	

02/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.87	3.75	59.68	350.49	960,000.00 *gal*
Application event totals		46.87	3.75	59.68	350.49	

02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.87	3.75	59.68	350.49	960,000.00 *gal*
Application event totals		46.87	3.75	59.68	350.49	

03/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.07	2.73	46.34	225.32	720,000.00 *gal*
Application event totals		29.07	2.73	46.34	225.32	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.07	2.73	46.34	225.32	720,000.00 *gal*
Application event totals		29.07	2.73	46.34	225.32	

#1 - 06/05/2023: Corn, silage

Field name: #1

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.57	17.88	84.49	405.57	3,240,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.64	0.00	0.00	117.73	5,940,000.00 *gal*
Application event totals		67.20	17.88	84.49	523.29	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#1 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,728,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,168,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#11 - 10/12/2022: Oats, silage-soft dough

Field name: #11

Crop: Oats, silage-soft dough

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.32	2.58	41.16	241.72	240,000.00 *gal*
Application event totals		32.32	2.58	41.16	241.72	

#11 - 10/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.32	2.58	41.16	241.72	240,000.00 *gal*
Application event totals		32.32	2.58	41.16	241.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.73	2.51	42.61	207.19	240,000.00 *gal*
Application event totals		26.73	2.51	42.61	207.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.73	2.51	42.61	207.19	240,000.00 *gal*
Application event totals		26.73	2.51	42.61	207.19	

#11 - 06/14/2023: Corn, silage

Field name: #11

Crop: Corn, silage

Plant date: 06/14/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.99	18.27	86.33	414.37	1,200,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.64	0.00	0.00	118.10	2,160,000.00 *gal*
Application event totals		68.63	18.27	86.33	532.47	

#11 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#11 - 06/14/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#12 - 10/12/2022: Oats, silage-soft dough

Field name: #12

Crop: Oats, silage-soft dough

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.24	2.50	39.79	233.66	120,000.00 *gal*
Application event totals		31.24	2.50	39.79	233.66	

#12 - 10/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.24	2.50	39.79	233.66	120,000.00 *gal*
Application event totals		31.24	2.50	39.79	233.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.24	2.50	39.79	233.66	120,000.00 *gal*
Application event totals		31.24	2.50	39.79	233.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.84	2.43	41.19	200.28	120,000.00 *gal*
Application event totals		25.84	2.43	41.19	200.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.84	2.43	41.19	200.28	120,000.00 *gal*
Application event totals		25.84	2.43	41.19	200.28	

#12 - 06/16/2023: Corn, silage

Field name: #12

Crop: Corn, silage

Plant date: 06/16/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.76	17.66	83.45	400.56	600,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.64	0.00	0.00	117.96	1,116,000.00 *gal*
Application event totals		66.40	17.66	83.45	518.52	

#12 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	324,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	324,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	324,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/28/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	324,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	324,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#12 - 06/16/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	324,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.90	7.06	33.38	160.22	240,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	594,000.00 *gal*
Application event totals		26.78	7.06	33.38	223.01	

#2 - 10/10/2022: Oats, silage-soft dough

Field name: #2

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.10	3.68	58.71	344.74	720,000.00 *gal*
Application event totals		46.10	3.68	58.71	344.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.10	3.68	58.71	344.74	720,000.00 *gal*
Application event totals		46.10	3.68	58.71	344.74	

#2 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	46.10	3.68	58.71	344.74	720,000.00 *gal*
Application event totals		46.10	3.68	58.71	344.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.12	3.59	60.77	295.50	720,000.00 *gal*
Application event totals		38.12	3.59	60.77	295.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	31.77	2.99	50.64	246.25	600,000.00 *gal*
Application event totals		31.77	2.99	50.64	246.25	

#2 - 06/06/2023: Corn, silage

Field name: #2

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.92	17.98	84.95	407.78	2,484,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.64	0.00	0.00	118.37	4,554,000.00 *gal*
Application event totals		67.57	17.98	84.95	526.15	

#2 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.03	9.55	45.15	216.70	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		35.88	9.55	45.15	277.52	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.03	9.55	45.15	216.70	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		35.88	9.55	45.15	277.52	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.03	9.55	45.15	216.70	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		35.88	9.55	45.15	277.52	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.03	9.55	45.15	216.70	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		35.88	9.55	45.15	277.52	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.03	9.55	45.15	216.70	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		35.88	9.55	45.15	277.52	

#2 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.03	9.55	45.15	216.70	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		35.88	9.55	45.15	277.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well/ Deep Well Dairy	Ground water	0.85	0.00	0.00	60.82	2,340,000.00 *gal*
Application event totals		0.85	0.00	0.00	60.82	

#3 - 10/10/2022: Oats, silage-soft dough

Field name: #3

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.88	2.71	43.14	253.37	720,000.00 *gal*
Application event totals		33.88	2.71	43.14	253.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.88	2.71	43.14	253.37	720,000.00 *gal*
Application event totals		33.88	2.71	43.14	253.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.88	2.71	43.14	253.37	720,000.00 *gal*
Application event totals		33.88	2.71	43.14	253.37	

#3 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.02	2.64	44.66	217.17	720,000.00 *gal*
Application event totals		28.02	2.64	44.66	217.17	

03/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.35	2.20	37.22	180.98	600,000.00 *gal*
Application event totals		23.35	2.20	37.22	180.98	

#3 - 06/07/2023: Corn, silage

Field name: #3

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.20	17.23	81.44	390.91	3,240,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.62	0.00	0.00	116.91	6,120,000.00 *gal*
Application event totals		64.82	17.23	81.44	507.82	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.11	9.57	45.24	217.17	1,800,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		35.98	9.57	45.24	279.96	

#3 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.11	9.57	45.24	217.17	1,800,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		35.98	9.57	45.24	279.96	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		0.87	0.00	0.00	62.79	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.11	9.57	45.24	217.17	1,800,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		35.98	9.57	45.24	279.96	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.11	9.57	45.24	217.17	1,800,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		35.98	9.57	45.24	279.96	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.09	7.66	36.20	173.74	1,440,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		28.96	7.66	36.20	236.52	

#3 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	3,286,800.00 *gal*
Application event totals		0.87	0.00	0.00	62.79	

#4 - 10/10/2022: Oats, silage-soft dough

Field name: #4

Crop: Oats, silage-soft dough Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.61	3.40	54.26	318.63	360,000.00 *gal*
Application event totals		42.61	3.40	54.26	318.63	

02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.61	3.40	54.26	318.63	360,000.00 *gal*
Application event totals		42.61	3.40	54.26	318.63	

02/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	42.61	3.40	54.26	318.63	360,000.00 *gal*
Application event totals		42.61	3.40	54.26	318.63	

03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.23	3.31	56.17	273.11	360,000.00 *gal*
Application event totals		35.23	3.31	56.17	273.11	

#4 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.23	3.31	56.17	273.11	360,000.00 *gal*
Application event totals		35.23	3.31	56.17	273.11	

#4 - 06/08/2023: Corn, silage

Field name: #4

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.76	17.66	83.45	400.56	1,320,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.68	0.00	0.00	121.08	2,520,000.00 *gal*
Application event totals		66.44	17.66	83.45	521.64	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.32	9.63	45.52	218.49	720,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		36.19	9.63	45.52	281.28	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.32	9.63	45.52	218.49	720,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		36.19	9.63	45.52	281.28	

#4 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.32	9.63	45.52	218.49	720,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		36.19	9.63	45.52	281.28	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.32	9.63	45.52	218.49	720,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		36.19	9.63	45.52	281.28	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	35.32	9.63	45.52	218.49	720,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		36.19	9.63	45.52	281.28	

08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.44	8.03	37.93	182.07	600,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		30.31	8.03	37.93	244.86	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.44	8.03	37.93	182.07	600,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,306,800.00 *gal*
Application event totals		30.31	8.03	37.93	244.86	

#5 - 10/11/2022: Oats, silage-soft dough

#5 - 10/11/2022: Oats, silage-soft dough

Field name: #5

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.40	2.27	36.17	212.42	240,000.00 *gal*
Application event totals		28.40	2.27	36.17	212.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.40	2.27	36.17	212.42	240,000.00 *gal*
Application event totals		28.40	2.27	36.17	212.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.49	2.21	37.45	182.07	240,000.00 *gal*
Application event totals		23.49	2.21	37.45	182.07	

#5 - 06/09/2023: Corn, silage

Field name: #5

Crop: Corn, silage

Plant date: 06/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.38	17.83	84.25	404.41	2,100,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.64	0.00	0.00	118.00	3,870,000.00 *gal*
Application event totals		67.02	17.83	84.25	522.41	

#5 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,123,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,123,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,123,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,123,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,123,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#5 - 06/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,123,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	14.94	4.07	19.26	92.44	480,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,059,200.00 *gal*
Application event totals		15.82	4.07	19.26	155.22	

#6 - 10/11/2022: Oats, silage-soft dough

Field name: #6

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.63	3.57	56.84	333.80	240,000.00 *gal*
Application event totals		44.63	3.57	56.84	333.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.63	3.57	56.84	333.80	240,000.00 *gal*
Application event totals		44.63	3.57	56.84	333.80	

#6 - 10/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	44.63	3.57	56.84	333.80	240,000.00 *gal*
Application event totals		44.63	3.57	56.84	333.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.91	3.47	58.84	286.11	240,000.00 *gal*
Application event totals		36.91	3.47	58.84	286.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	36.91	3.47	58.84	286.11	240,000.00 *gal*
Application event totals		36.91	3.47	58.84	286.11	

#6 - 06/10/2023: Corn, silage

Field name: #6

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.76	17.66	83.45	400.56	840,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.70	0.00	0.00	122.31	1,620,000.00 *gal*
Application event totals		66.46	17.66	83.45	522.87	

#6 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#6 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	453,600.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	831,600.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#7 - 10/11/2022: Oats, silage-soft dough

Field name: #7

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.68	1.81	28.88	169.59	360,000.00 *gal*
Application event totals		22.68	1.81	28.88	169.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.68	1.81	28.88	169.59	360,000.00 *gal*
Application event totals		22.68	1.81	28.88	169.59	

#7 - 10/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.68	1.81	28.88	169.59	360,000.00 *gal*
Application event totals		22.68	1.81	28.88	169.59	

03/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	18.75	1.76	29.90	145.36	360,000.00 *gal*
Application event totals		18.75	1.76	29.90	145.36	

#7 - 06/11/2023: Corn, silage

Field name: #7

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.80	17.94	84.80	407.02	2,520,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.66	0.00	0.00	119.68	4,680,000.00 *gal*
Application event totals		67.46	17.94	84.80	526.70	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,339,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,339,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,339,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,339,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,339,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	1,339,200.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#7 - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	2,455,200.00 *gal*
Application event totals		0.87	0.00	0.00	62.79	

#8 - 10/11/2022: Oats, silage-soft dough

Field name: #8

Crop: Oats, silage-soft dough

Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.48	3.87	61.74	362.58	360,000.00 *gal*
Application event totals		48.48	3.87	61.74	362.58	

02/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.48	3.87	61.74	362.58	360,000.00 *gal*
Application event totals		48.48	3.87	61.74	362.58	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	48.48	3.87	61.74	362.58	360,000.00 *gal*
Application event totals		48.48	3.87	61.74	362.58	

03/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.09	3.77	63.92	310.78	360,000.00 *gal*
Application event totals		40.09	3.77	63.92	310.78	

#8 - 10/11/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	40.09	3.77	63.92	310.78	360,000.00 *gal*
Application event totals		40.09	3.77	63.92	310.78	

#8 - 06/12/2023: Corn, silage

Field name: #8

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.99	18.27	86.33	414.37	1,200,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.64	0.00	0.00	118.10	2,160,000.00 *gal*
Application event totals		68.63	18.27	86.33	532.47	

07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#8 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	626,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	23.45	6.39	30.21	145.03	420,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		24.32	6.39	30.21	207.82	

08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	1,148,400.00 *gal*
Application event totals		0.87	0.00	0.00	62.79	

#9 - 10/12/2022: Oats, silage-soft dough

#9 - 10/12/2022: Oats, silage-soft dough

Field name: #9

Crop: Oats, silage-soft dough

Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.05	3.12	49.74	292.08	240,000.00 *gal*
Application event totals		39.05	3.12	49.74	292.08	

02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.05	3.12	49.74	292.08	240,000.00 *gal*
Application event totals		39.05	3.12	49.74	292.08	

02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	39.05	3.12	49.74	292.08	240,000.00 *gal*
Application event totals		39.05	3.12	49.74	292.08	

03/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.30	3.04	51.49	250.35	240,000.00 *gal*
Application event totals		32.30	3.04	51.49	250.35	

03/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	32.30	3.04	51.49	250.35	240,000.00 *gal*
Application event totals		32.30	3.04	51.49	250.35	

#9 - 06/13/2023: Corn, silage

Field name: #9

Crop: Corn, silage

Plant date: 06/13/2023

#9 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.76	17.66	83.45	400.56	960,000.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	1.65	0.00	0.00	118.92	1,800,000.00 *gal*
Application event totals		66.41	17.66	83.45	519.48	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water518,400.00	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

#9 - 06/13/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

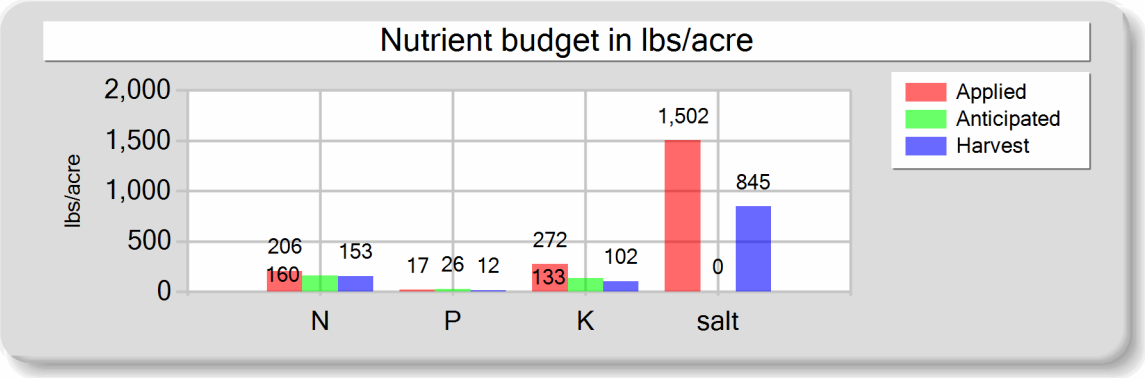
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	34.97	9.54	45.06	216.30	518,400.00 *gal*
Ag Well/ Deep Well Dairy	Ground water	0.87	0.00	0.00	62.79	950,400.00 *gal*
Application event totals		35.84	9.54	45.06	279.09	

B. NUTRIENT BUDGET

#1 - 10/10/2022: Oats, silage-soft dough

Field name: #1 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.73	16.70	271.73	1,502.10
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.73	16.70	271.73	1,502.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	153.28	11.82	102.49	844.87
Nutrient balance	52.45	4.88	169.24	657.23
Applied to removed ratio	1.34	1.41	2.65	1.78

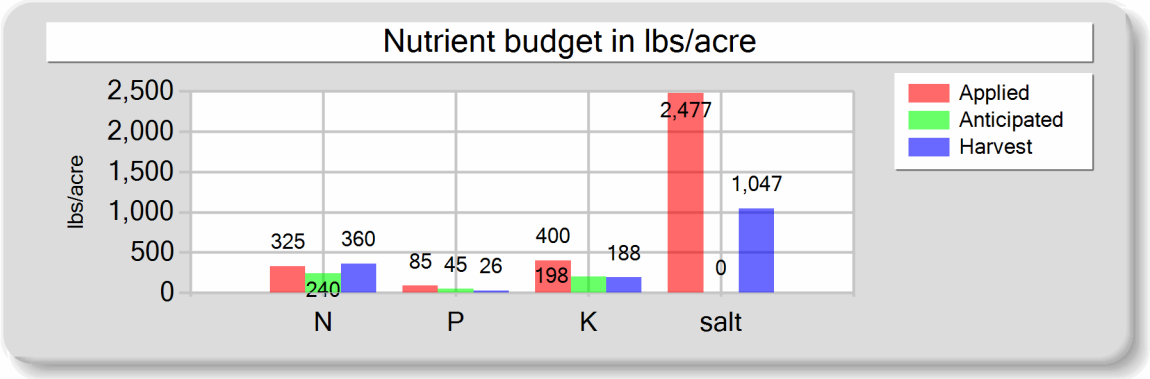
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
4,320,000.00 *gallons*
159.09 *acre-inches*
1.99 *inches/acre*

Total harvests for the crop
1 *harvests*

#1 - 06/05/2023: Corn, silage

Field name: #1 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	310.35	84.63	399.93	1,919.68
Fresh water	7.74	0.00	0.00	557.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.09	84.63	399.93	2,476.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	359.97	26.11	188.09	1,047.49
Nutrient balance	-34.88	58.52	211.84	1,429.44
Applied to removed ratio	0.90	3.24	2.13	2.36

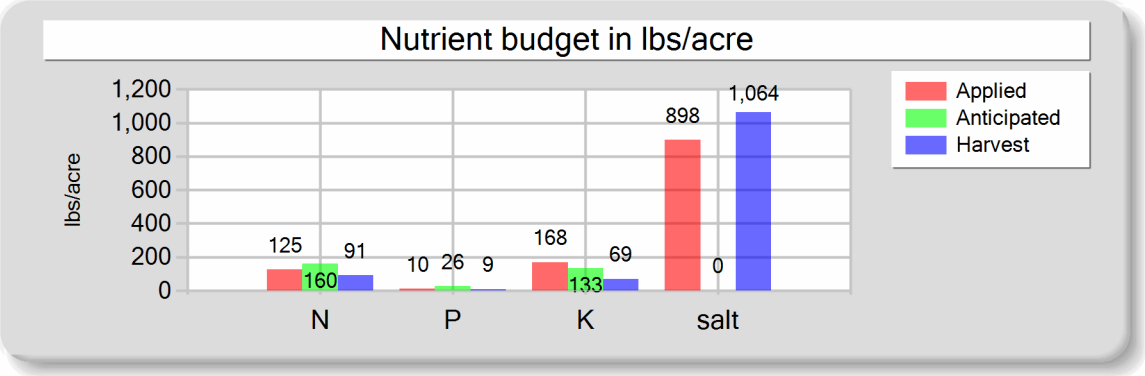
Fresh water applied	
28,116,000.00	*gallons*
1,035.42	*acre-inches*
12.94	*inches/acre*

Process wastewater applied	
15,336,000.00	*gallons*
564.77	*acre-inches*
7.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

#11 - 10/12/2022: Oats, silage-soft dough

Field name: #11 Crop: Oats, silage-soft dough Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	118.10	10.19	167.54	897.81
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	125.10	10.19	167.54	897.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	90.67	9.07	69.47	1,064.09
Nutrient balance	34.42	1.13	98.08	-166.28
Applied to removed ratio	1.38	1.12	2.41	0.84

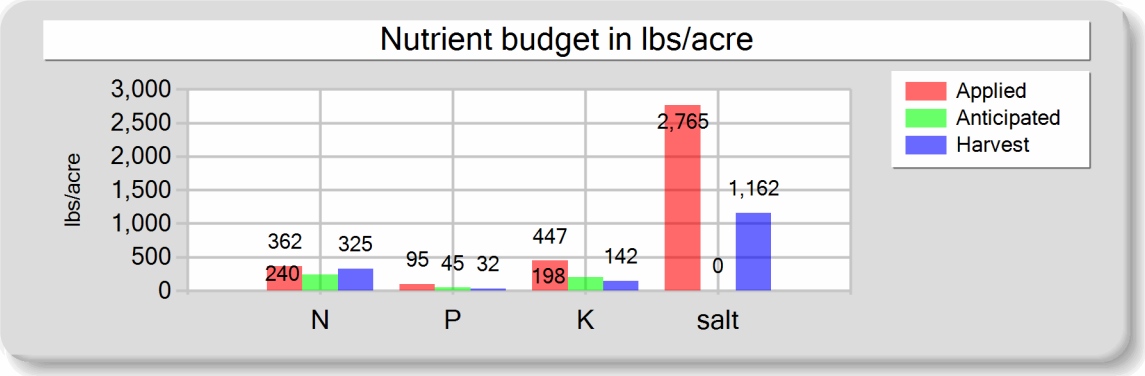
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
960,000.00 *gallons*
35.35 *acre-inches*
1.22 *inches/acre*

Total harvests for the crop
1 *harvests*

#11 - 06/14/2023: Corn, silage

Field name: #11 Crop: Corn, silage Plant date: 06/14/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	346.74	94.55	446.83	2,144.79
Fresh water	8.62	0.00	0.00	620.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	362.36	94.55	446.83	2,765.19
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	325.12	31.98	142.48	1,161.90
Nutrient balance	37.24	62.57	304.35	1,603.29
Applied to removed ratio	1.11	2.96	3.14	2.38

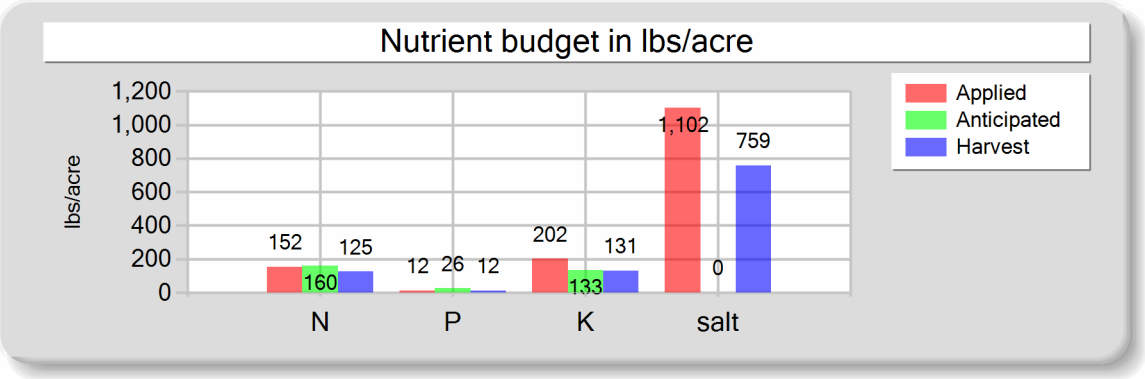
Fresh water applied	
11,347,200.00	*gallons*
417.88	*acre-inches*
14.41	*inches/acre*

Process wastewater applied	
6,211,200.00	*gallons*
228.74	*acre-inches*
7.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

#12 - 10/12/2022: Oats, silage-soft dough

Field name: #12 Crop: Oats, silage-soft dough Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	145.40	12.35	201.75	1,101.54
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	152.40	12.35	201.75	1,101.54
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	125.23	12.16	130.71	758.69
Nutrient balance	27.17	0.19	71.04	342.85
Applied to removed ratio	1.22	1.02	1.54	1.45

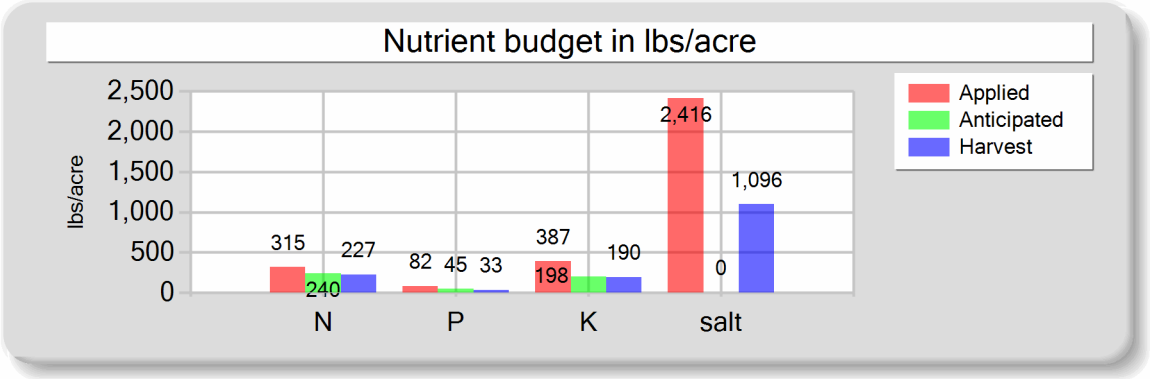
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
600,000.00	*gallons*
22.10	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

#12 - 06/16/2023: Corn, silage

Field name: #12 Crop: Corn, silage Plant date: 06/16/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	300.47	81.93	387.21	1,858.60
Fresh water	7.75	0.00	0.00	557.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	315.22	81.93	387.21	2,416.08
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	226.56	32.97	190.25	1,096.49
Nutrient balance	88.66	48.97	196.96	1,319.59
Applied to removed ratio	1.39	2.49	2.04	2.20

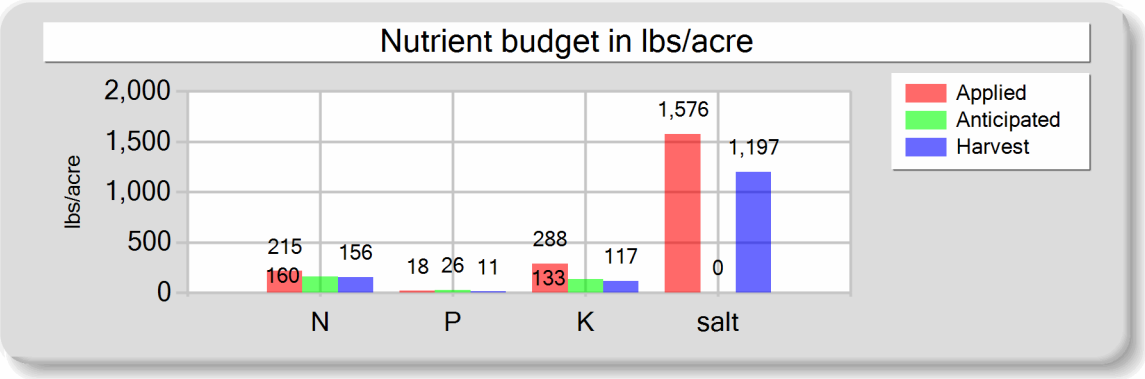
Fresh water applied	
5,274,000.00	*gallons*
194.22	*acre-inches*
12.95	*inches/acre*

Process wastewater applied	
2,784,000.00	*gallons*
102.53	*acre-inches*
6.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 10/10/2022: Oats, silage-soft dough

Field name: #2 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	208.18	17.62	287.53	1,575.97
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	215.18	17.62	287.53	1,575.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	155.51	10.61	117.25	1,197.21
Nutrient balance	59.66	7.01	170.28	378.76
Applied to removed ratio	1.38	1.66	2.45	1.32

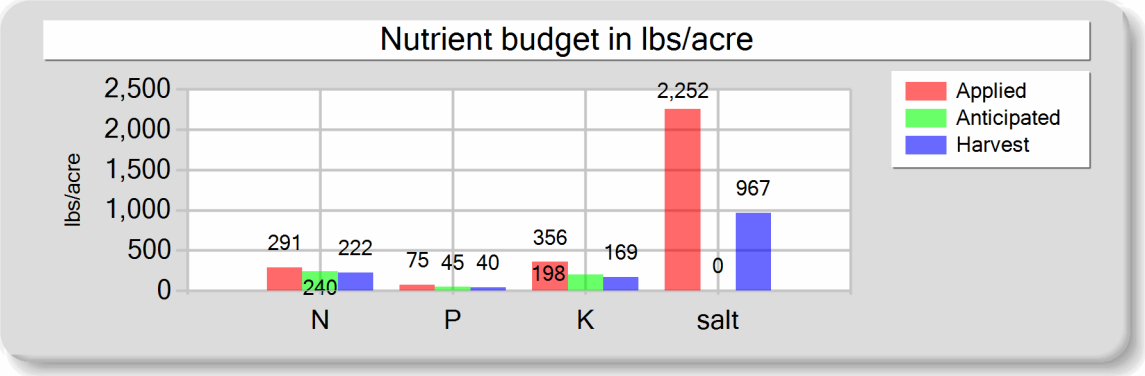
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
3,480,000.00 *gallons*
128.16 *acre-inches*
2.10 *inches/acre*

Total harvests for the crop
1 *harvests*

#2 - 06/06/2023: Corn, silage

Field name: #2 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	276.12	75.29	355.83	1,707.96
Fresh water	7.56	0.00	0.00	544.13
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.68	75.29	355.83	2,252.09
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	222.47	40.46	169.27	967.26
Nutrient balance	68.21	34.83	186.55	1,284.83
Applied to removed ratio	1.31	1.86	2.10	2.33

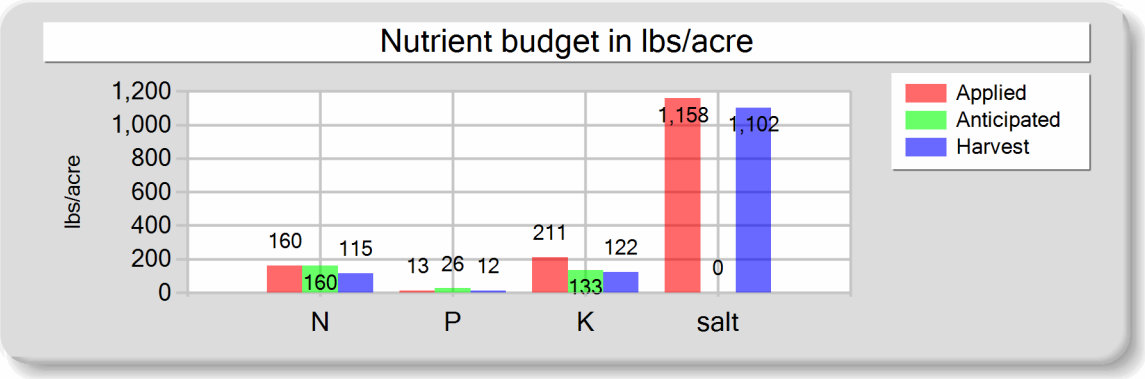
Fresh water applied	
20,934,000.00	*gallons*
770.93	*acre-inches*
12.64	*inches/acre*

Process wastewater applied	
10,404,000.00	*gallons*
383.14	*acre-inches*
6.28	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 10/10/2022: Oats, silage-soft dough

Field name: #3 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	153.00	12.95	211.32	1,158.25
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	160.00	12.95	211.32	1,158.25
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	115.34	11.96	122.22	1,101.72
Nutrient balance	44.66	0.99	89.10	56.52
Applied to removed ratio	1.39	1.08	1.73	1.05

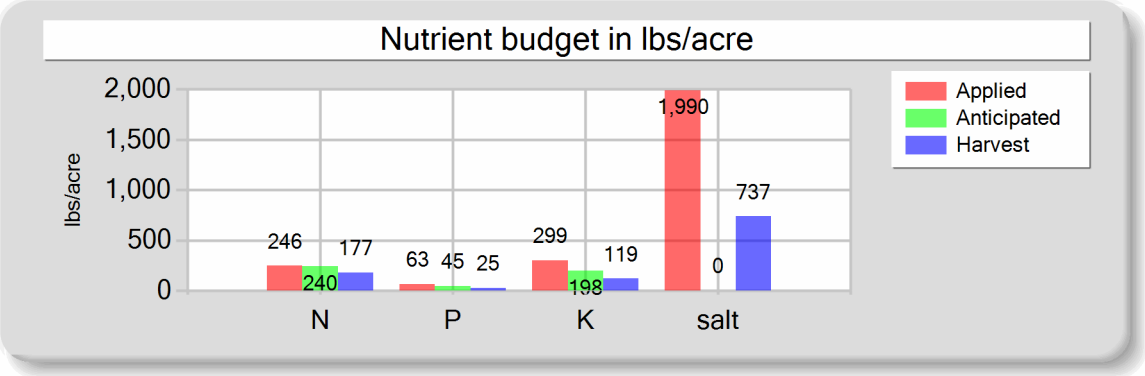
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
3,480,000.00 *gallons*
128.16 *acre-inches*
1.54 *inches/acre*

Total harvests for the crop
1 *harvests*

#3 - 06/07/2023: Corn, silage

Field name: #3 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	231.72	63.19	298.61	1,433.33
Fresh water	7.73	0.00	0.00	556.42
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	246.45	63.19	298.61	1,989.75
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	177.48	24.82	119.18	737.32
Nutrient balance	68.98	38.37	179.43	1,252.43
Applied to removed ratio	1.39	2.55	2.51	2.70

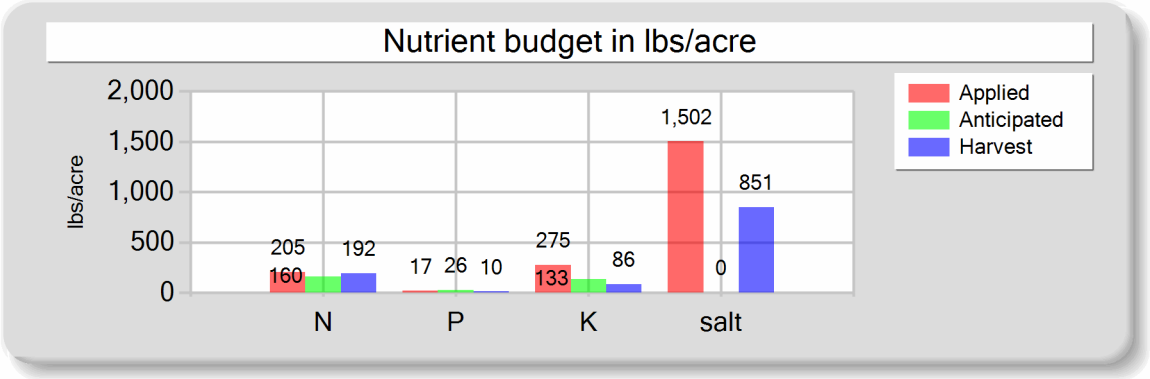
Fresh water applied	
29,127,600.00	*gallons*
1,072.67	*acre-inches*
12.92	*inches/acre*

Process wastewater applied	
11,880,000.00	*gallons*
437.50	*acre-inches*
5.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 10/10/2022: Oats, silage-soft dough

Field name: #4 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.28	16.84	275.11	1,502.10
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.28	16.84	275.11	1,502.10
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	192.49	9.91	85.50	850.65
Nutrient balance	12.78	6.93	189.61	651.45
Applied to removed ratio	1.07	1.70	3.22	1.77

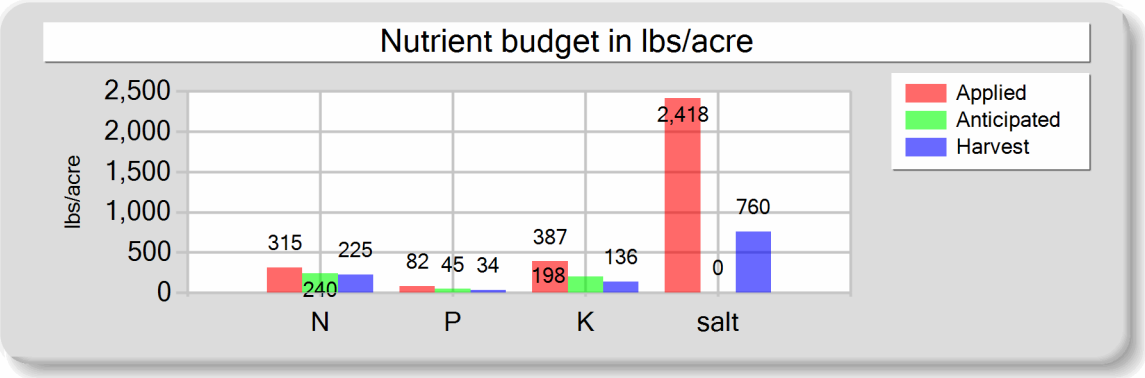
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
2.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 06/08/2023: Corn, silage

Field name: #4 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	300.24	81.87	386.90	1,857.14
Fresh water	7.79	0.00	0.00	560.59
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	315.03	81.87	386.90	2,417.73
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	224.91	34.05	136.22	760.27
Nutrient balance	90.11	47.82	250.69	1,657.46
Applied to removed ratio	1.40	2.40	2.84	3.18

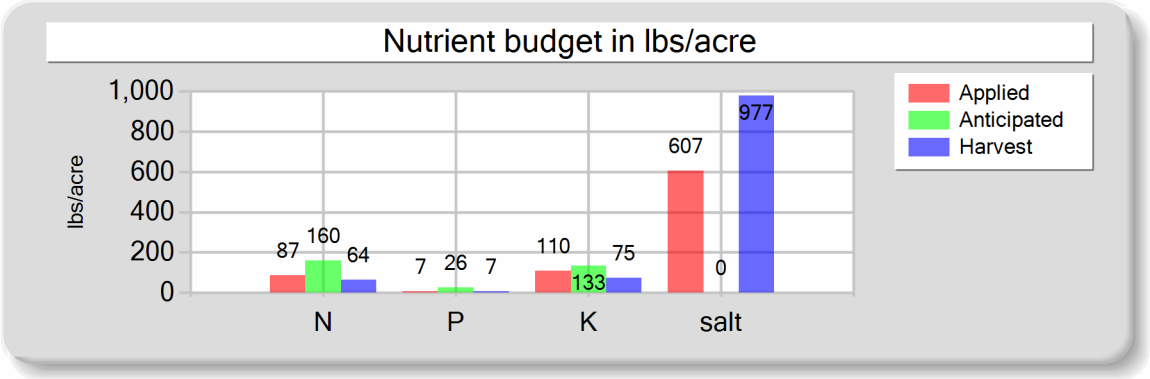
Fresh water applied	
11,667,600.00	*gallons*
429.68	*acre-inches*
13.02	*inches/acre*

Process wastewater applied	
6,120,000.00	*gallons*
225.38	*acre-inches*
6.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

#5 - 10/11/2022: Oats, silage-soft dough

Field name: #5 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	80.29	6.75	109.79	606.91
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	87.29	6.75	109.79	606.91
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	63.77	6.82	74.56	977.18
Nutrient balance	23.52	-0.07	35.23	-370.27
Applied to removed ratio	1.37	0.99	1.47	0.62

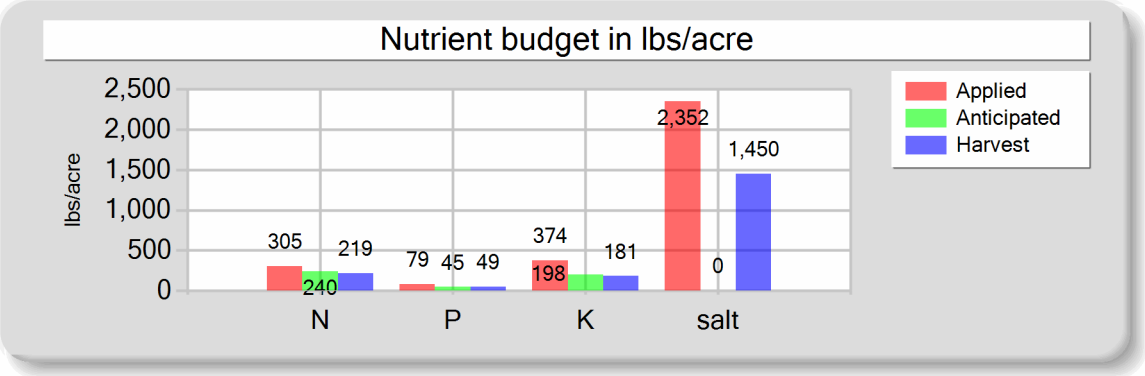
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
720,000.00	*gallons*
26.52	*acre-inches*
0.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

#5 - 06/09/2023: Corn, silage

Field name: #5 Crop: Corn, silage Plant date: 06/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.14	79.11	373.89	1,794.66
Fresh water	7.75	0.00	0.00	557.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.88	79.11	373.89	2,352.18
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	219.00	48.61	180.84	1,450.05
Nutrient balance	85.88	30.50	193.05	902.13
Applied to removed ratio	1.39	1.63	2.07	1.62

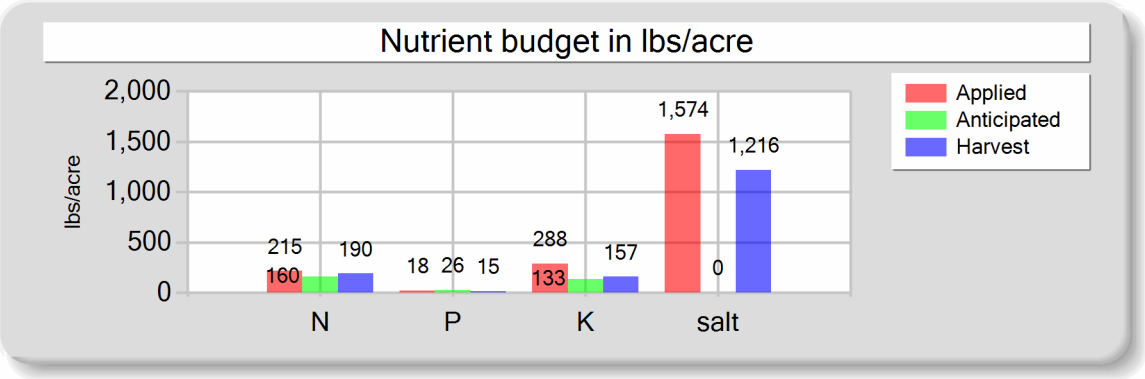
Fresh water applied	
18,284,400.00	*gallons*
673.35	*acre-inches*
12.95	*inches/acre*

Process wastewater applied	
9,319,200.00	*gallons*
343.19	*acre-inches*
6.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

#6 - 10/11/2022: Oats, silage-soft dough

Field name: #6 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.72	17.64	288.21	1,573.63
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.72	17.64	288.21	1,573.63
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	189.65	14.69	157.07	1,216.05
Nutrient balance	25.07	2.95	131.14	357.58
Applied to removed ratio	1.13	1.20	1.83	1.29

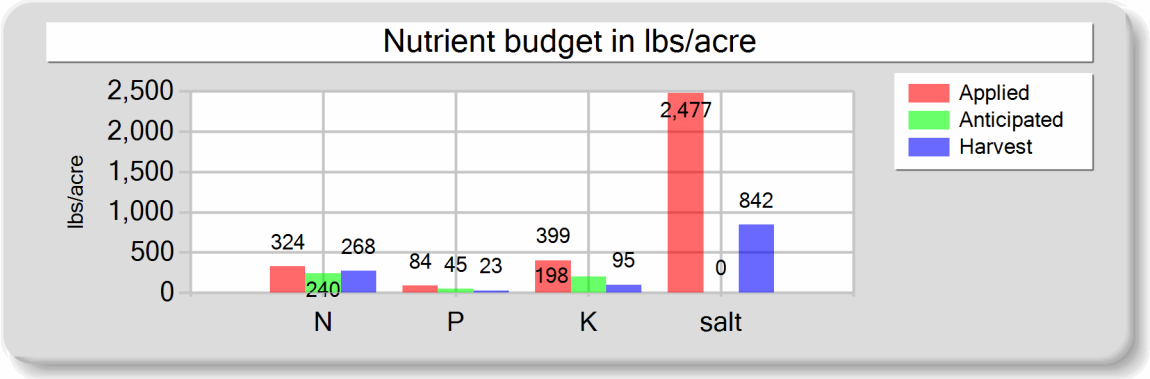
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
1,200,000.00 *gallons*
44.19 *acre-inches*
2.10 *inches/acre*

Total harvests for the crop
1 *harvests*

#6 - 06/10/2023: Corn, silage

Field name: #6 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	309.54	84.41	398.89	1,914.68
Fresh water	7.81	0.00	0.00	561.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.35	84.41	398.89	2,476.51
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	268.24	22.97	95.27	841.71
Nutrient balance	56.11	61.44	303.62	1,634.79
Applied to removed ratio	1.21	3.67	4.19	2.94

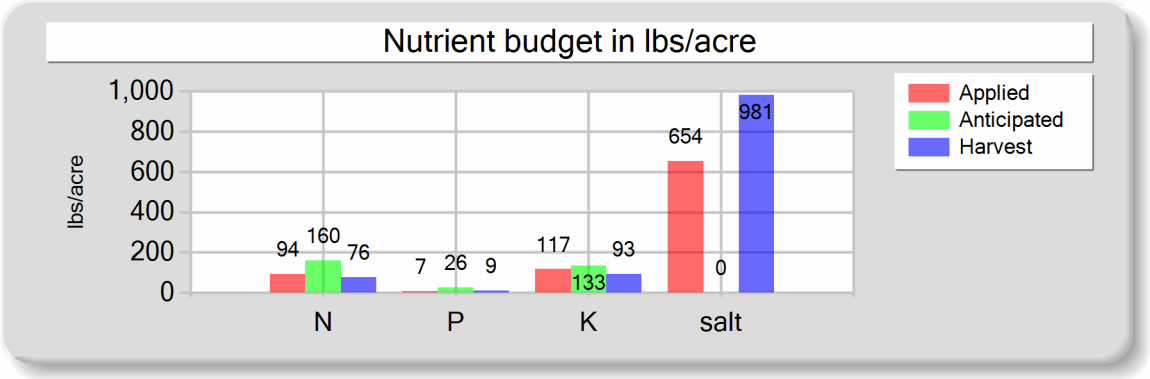
Fresh water applied	
7,441,200.00	*gallons*
274.03	*acre-inches*
13.05	*inches/acre*

Process wastewater applied	
4,015,200.00	*gallons*
147.87	*acre-inches*
7.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

#7 - 10/11/2022: Oats, silage-soft dough

Field name: #7 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	86.78	7.20	116.53	654.14
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	93.78	7.20	116.53	654.14
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	76.48	8.87	93.08	981.27
Nutrient balance	17.30	-1.67	23.46	-327.13
Applied to removed ratio	1.23	0.81	1.25	0.67

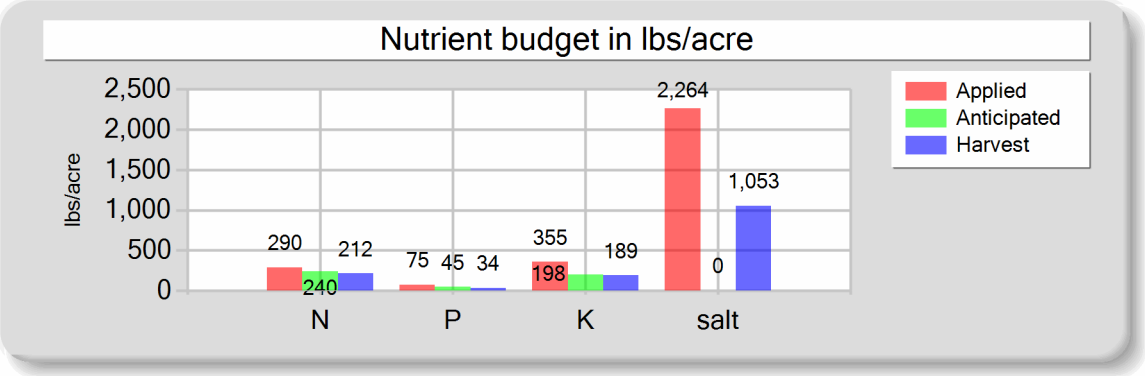
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,440,000.00	*gallons*
53.03	*acre-inches*
0.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

#7 - 06/11/2023: Corn, silage

Field name: #7 Crop: Corn, silage Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	275.61	75.15	355.17	1,704.84
Fresh water	7.77	0.00	0.00	559.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.38	75.15	355.17	2,264.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	212.04	34.42	188.80	1,052.92
Nutrient balance	78.35	40.74	166.37	1,211.12
Applied to removed ratio	1.37	2.18	1.88	2.15

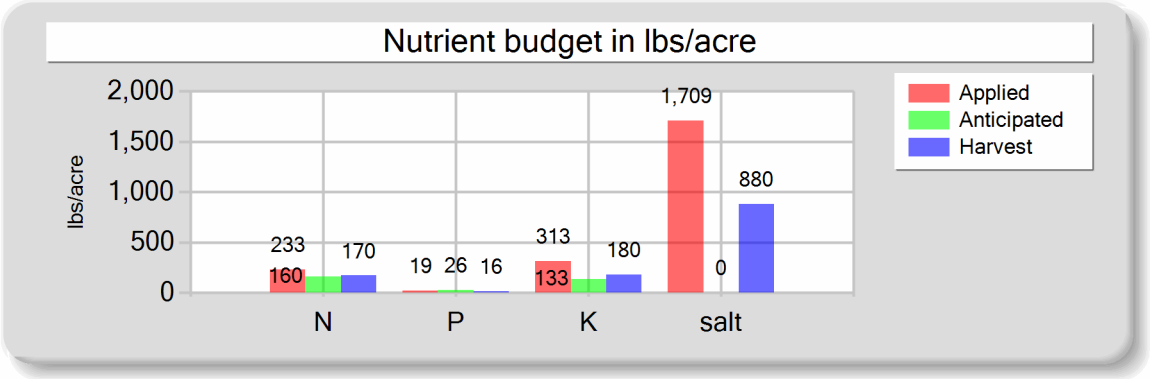
Fresh water applied	
21,866,400.00	*gallons*
805.27	*acre-inches*
12.99	*inches/acre*

Process wastewater applied	
10,555,200.00	*gallons*
388.71	*acre-inches*
6.27	*inches/acre*

Total harvests for the crop	
1	*harvests*

#8 - 10/11/2022: Oats, silage-soft dough

Field name: #8 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.63	19.16	313.06	1,709.29
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	232.63	19.16	313.06	1,709.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	170.21	15.99	180.04	879.51
Nutrient balance	62.41	3.18	133.02	829.77
Applied to removed ratio	1.37	1.20	1.74	1.94

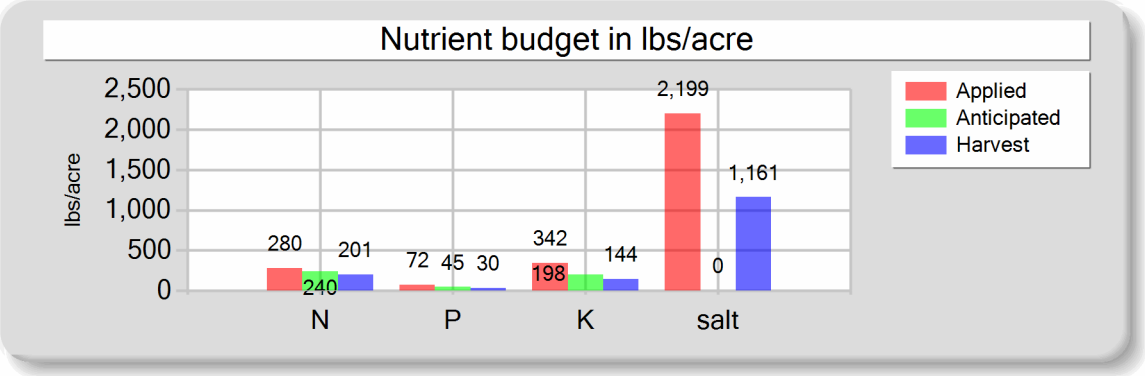
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,800,000.00	*gallons*
66.29	*acre-inches*
2.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

#8 - 06/12/2023: Corn, silage

Field name: #8 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	265.28	72.34	341.86	1,640.91
Fresh water	7.75	0.00	0.00	557.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.03	72.34	341.86	2,198.53
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	200.53	30.34	143.86	1,161.30
Nutrient balance	79.50	42.00	198.00	1,037.22
Applied to removed ratio	1.40	2.38	2.38	1.89

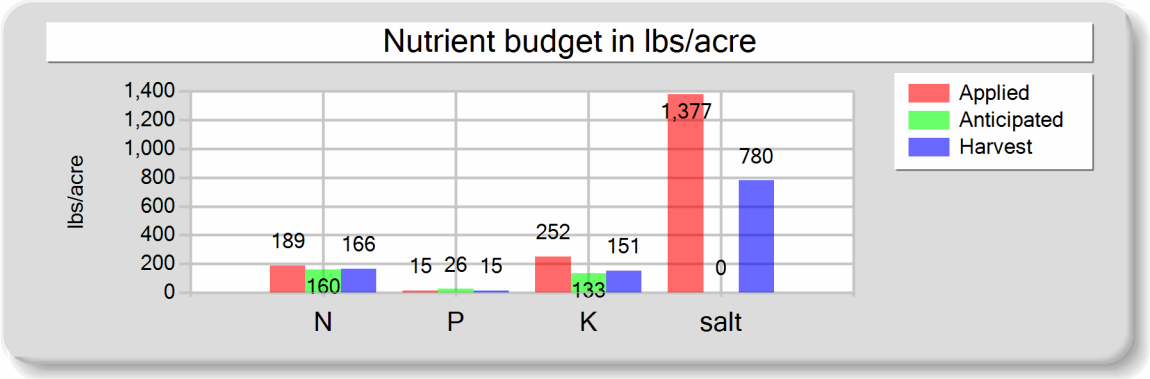
Fresh water applied	
10,198,800.00	*gallons*
375.59	*acre-inches*
12.95	*inches/acre*

Process wastewater applied	
4,752,000.00	*gallons*
175.00	*acre-inches*
6.03	*inches/acre*

Total harvests for the crop	
1	*harvests*

#9 - 10/12/2022: Oats, silage-soft dough

Field name: #9 Crop: Oats, silage-soft dough Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.75	15.44	252.19	1,376.93
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.75	15.44	252.19	1,376.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	166.14	15.01	151.03	779.52
Nutrient balance	22.62	0.43	101.15	597.41
Applied to removed ratio	1.14	1.03	1.67	1.77

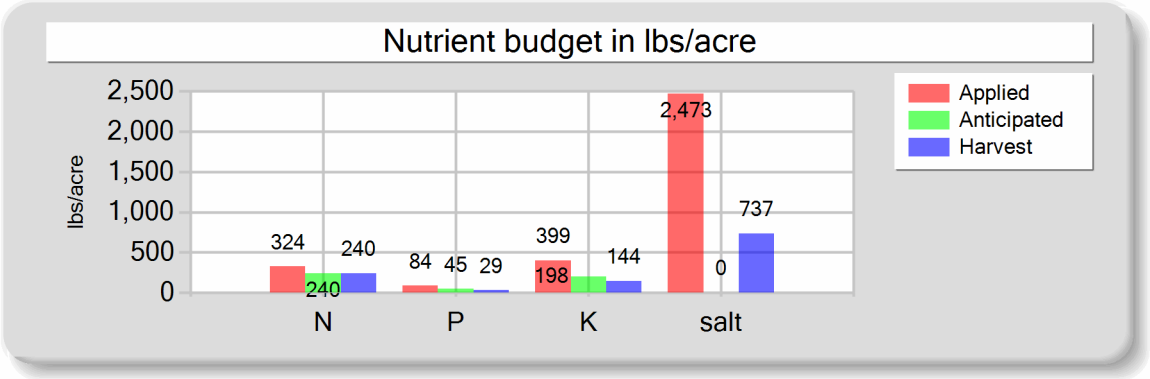
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
1,200,000.00	*gallons*
44.19	*acre-inches*
1.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

#9 - 06/13/2023: Corn, silage

Field name: #9 Crop: Corn, silage Plant date: 06/13/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	309.54	84.41	398.89	1,914.68
Fresh water	7.76	0.00	0.00	558.43
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	324.30	84.41	398.89	2,473.11
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	239.72	28.50	143.54	736.70
Nutrient balance	84.58	55.90	255.35	1,736.41
Applied to removed ratio	1.35	2.96	2.78	3.36

Fresh water applied	
8,452,800.00	*gallons*
311.29	*acre-inches*
12.97	*inches/acre*

Process wastewater applied	
4,588,800.00	*gallons*
168.99	*acre-inches*
7.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2102514-01

Sample and source description: Manure 2102514-01

Sample date: 10/25/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,460.00	4,300.00	19,400.00	7,500.00	4,200.00	2,700.00	3,310.00	11,200.00		34.30
DL	0.50	0.50	0.50	5.00	5.00	5.00	1.00	0.50		0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2302041-01

Sample and source description: Lagoon Water 2302041-01

Sample date: 01/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	468.00	468.00			37.40	596.00								4,520.00	3,500
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2303403-01

Sample and source description: Lagoon Water 2303403-01

Sample date: 03/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	387.00	387.00			36.40	617.00								4,380.00	3,000
DL	0.60	5.00			0.10	10.00								1.00	1

Lagoon Water 2307248-01

Sample and source description: Lagoon Water 2307248-01

Sample date: 07/17/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	194.00	166.00			52.90	250.00								2,470.00	1,200
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310055-01

Sample and source description: Lagoon Water 2310055-01

Sample date: 09/29/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	472.00	352.00			104.00	573.00								4,780.00	2,500
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #10

2311136-02

Sample description: 2311136-02

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.73										264.00	190
DL	1.40										1.00	1

Ag Well #2 River Road

Ag Well #2 River Road

2311136-03

Sample description: 2311136-03

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.50										482.00	300
DL	1.40										1.00	1

Ag Well #3 River Road

2311136-04

Sample description: 2311136-04

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											401.00	260
DL											1.00	1

Ag Well #4 River Road

2311136-05

Sample description: 2311136-05

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.42										445.00	280
DL	1.40										1.00	1

Ag Well/ Deep Well Dairy

Ag Well/ Deep Well Dairy

2311136-01

Sample description: 2311136-01

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.64										263.00	190
DL	1.40										1.00	1

Domestic Well

2310490-01

Sample description: 2310490-01

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.78										245.00	
DL	0.40										1.00	

Reservoir SSJID

2311136-07

Sample description: 2311136-07

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.95										736.00	30
DL	0.40										1.00	1

Stanislaus River

Stanislaus River

2311136-06

Sample description: 2311136-06

Sample date: 11/12/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.06										74.00	40
DL	0.40										1.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

#1 - 10/10/2022: Oats, silage-soft dough

2305229-01

Sample and source description: 2305229-01

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,200.00	2,000.00	17,600.00		14.10
DL	0.50	0.50	0.50		0.10

2305229-02

Sample and source description: 2305229-02

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	33,200.00	2,560.00	22,200.00		18.30
DL	0.50	0.50	0.50		0.10

#1 - 10/10/2022: Oats, silage-soft dough

2305229-03

Sample and source description: 2305229-03

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,900.00	2,040.00	21,200.00		16.80
DL	0.50	0.50	0.50		0.10

#1 - 06/05/2023: Corn, silage

2310434-01

Sample and source description: 2310434-01

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	1,200.00	9,950.00		7.19
DL	0.50	0.50	0.50		0.10

2310434-02

Sample and source description: 2310434-02

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	1,610.00	11,600.00		6.46
DL	0.50	0.50	0.50		0.10

#11 - 10/12/2022: Oats, silage-soft dough

#11 - 10/12/2022: Oats, silage-soft dough

2305068-07

Sample and source description: 2305068-07

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	1,210.00	9,270.00		14.20
DL	2.50	0.50	0.50		0.10

#11 - 06/14/2023: Corn, silage

2310435-09

Sample and source description: 2310435-09

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,300.00	2,190.00	10,300.00		6.25
DL	0.50	0.50	0.50		0.10

2310435-10

Sample and source description: 2310435-10

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,300.00	1,800.00	8,020.00		6.54
DL	0.50	0.50	0.50		0.10

#12 - 10/12/2022: Oats, silage-soft dough

#12 - 10/12/2022: Oats, silage-soft dough

2305228-04

Sample and source description: 2305228-04

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 80.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,400.00	2,660.00	28,600.00		16.60
DL	0.50	0.50	0.50		0.10

#12 - 06/16/2023: Corn, silage

2310436-01

Sample and source description: 2310436-01

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	2,270.00	13,100.00		7.55
DL	0.50	0.50	0.50		0.10

2310436-02

Sample and source description: 2310436-02

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	1,930.00	11,200.00		7.97
DL	0.50	0.50	0.50		0.10

#2 - 10/10/2022: Oats, silage-soft dough

#2 - 10/10/2022: Oats, silage-soft dough

2305229-04

Sample and source description: 2305229-04

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 25.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,690.00	594.00	6,040.00		14.60
DL	0.50	0.50	0.50		0.10

2305229-05)

Sample and source description: 2305229-05)

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,200.00	1,720.00	19,000.00		19.40
DL	1.00	0.50	0.50		0.10

#2 - 06/06/2023: Corn, silage

2310434-03

Sample and source description: 2310434-03

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,510.00	10,500.00		6.00
DL	0.50	0.50	0.50		0.10

#2 - 06/06/2023: Corn, silage

2310434-04

Sample and source description: 2310434-04

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	2,210.00	8,510.00		5.39
DL	0.50	0.50	0.50		0.10

#3 - 10/10/2022: Oats, silage-soft dough

2305068-01

Sample and source description: 2305068-01

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	1,390.00	14,200.00		12.80
DL	2.50	0.50	0.50		0.10

2305068-02

Sample and source description: 2305068-02

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,790.00	1,310.00	15,300.00		11.90
DL	2.50	0.50	0.50		0.10

#3 - 06/07/2023: Corn, silage

#3 - 06/07/2023: Corn, silage

2310434-05

Sample and source description: 2310434-05

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	1,720.00	8,260.00		5.11
DL	0.50	0.50	0.50		0.10

2310434-06

Sample and source description: 2310434-06

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	1,400.00	6,370.00		4.26
DL	0.50	0.50	0.50		0.10

#4 - 10/10/2022: Oats, silage-soft dough

2305229-06

Sample and source description: 2305229-06

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 81.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	43,900.00	2,260.00	19,500.00		19.40
DL	2.50	0.50	0.50		0.10

#4 - 06/08/2023: Corn, silage

#4 - 06/08/2023: Corn, silage

2310434-07

Sample and source description: 2310434-07

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,200.00	2,150.00	8,600.00		4.80
DL	0.50	0.50	0.50		0.10

2310434-08

Sample and source description: 2310434-08

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	2,280.00	10,400.00		4.74
DL	0.50	0.50	0.50		0.10

#5 - 10/11/2022: Oats, silage-soft dough

2305068-03

Sample and source description: 2305068-03

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,100.00	637.00	7,510.00		11.10
DL	2.50	0.50	0.50		0.10

#5 - 10/11/2022: Oats, silage-soft dough

2305068-04)

Sample and source description: 2305068-04)

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,810.00	942.00	10,300.00		13.50
DL	2.50	0.50	0.50		0.10

#5 - 06/09/2023: Corn, silage

2310434-09

Sample and source description: 2310434-09

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,880.00	1,930.00	9,000.00		8.31
DL	0.50	0.50	0.50		0.10

2310434-10

Sample and source description: 2310434-10

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,930.00	10,900.00		8.74
DL	0.50	0.50	0.50		0.10

#6 - 10/11/2022: Oats, silage-soft dough

#6 - 10/11/2022: Oats, silage-soft dough

2305229-07

Sample and source description: 2305229-07

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,200.00	2,030.00	21,700.00		16.80
DL	0.50	0.50	0.50		0.10

#6 - 06/10/2023: Corn, silage

2310435-01

Sample and source description: 2310435-01

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	2,110.00	8,780.00		5.36
DL	0.50	0.50	0.50		0.10

2310435-02

Sample and source description: 2310435-02

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,400.00	1,490.00	6,180.00		5.46
DL	0.50	0.50	0.50		0.10

#7 - 10/11/2022: Oats, silage-soft dough

#7 - 10/11/2022: Oats, silage-soft dough

2305068-05

Sample and source description: 2305068-05

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	1,230.00	12,900.00		13.60
DL	2.50	0.50	0.50		0.10

2305068-06

Sample and source description: 2305068-06

Sample date: 05/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,340.00	871.00	10,400.00		16.80
DL	2.50	0.50	0.50		0.10

#7 - 06/11/2023: Corn, silage

2310435-03

Sample and source description: 2310435-03

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,110.00	11,000.00		6.34
DL	0.50	0.50	0.50		0.10

#7 - 06/11/2023: Corn, silage

2310435-04

Sample and source description: 2310435-04

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,370.00	13,000.00		7.25
DL	0.50	0.50	0.50		0.10

#8 - 10/11/2022: Oats, silage-soft dough

2305228-01

Sample and source description: 2305228-01

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 24.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,760.00	743.00	6,740.00		16.00
DL	1.00	0.50	0.50		0.10

2305228-02

Sample and source description: 2305228-02

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,900.00	3,090.00	34,800.00		17.00
DL	2.50	0.50	0.50		0.10

#8 - 06/12/2023: Corn, silage

#8 - 06/12/2023: Corn, silage

2310435-05

Sample and source description: 2310435-05

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	1,740.00	8,250.00		6.66
DL	0.50	0.50	0.50		0.10

2310435-06

Sample and source description: 2310435-06

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,590.00	2,270.00	11,600.00		5.29
DL	0.50	0.50	0.50		0.10

#9 - 10/12/2022: Oats, silage-soft dough

2305228-03

Sample and source description: 2305228-03

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 79.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,100.00	3,080.00	31,000.00		16.00
DL	5.00	0.50	0.50		0.10

#9 - 06/13/2023: Corn, silage

#9 - 06/13/2023: Corn, silage

2310435-07

Sample and source description: 2310435-07

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,600.00	2,850.00	16,100.00		5.30
DL	0.50	0.50	0.50		0.10

2310435-08

Sample and source description: 2310435-08

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	1,950.00	9,820.00		5.04
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

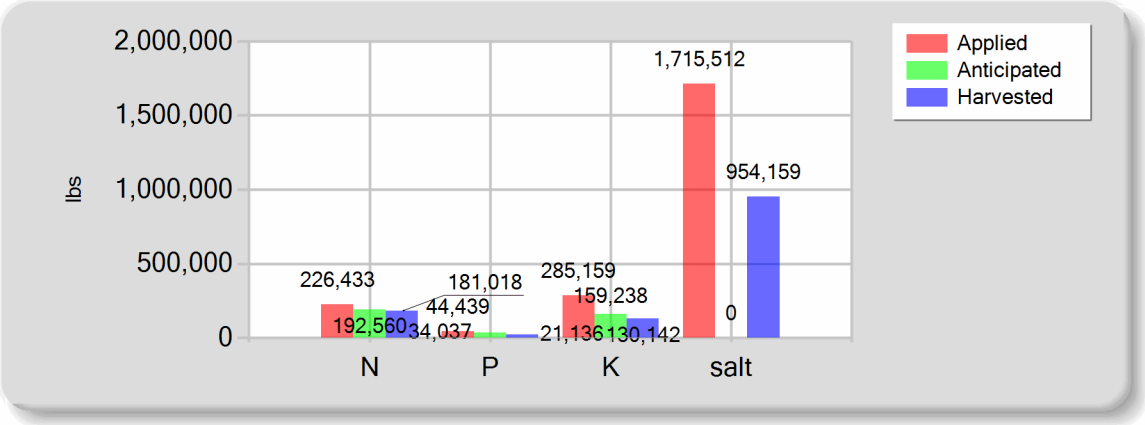
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

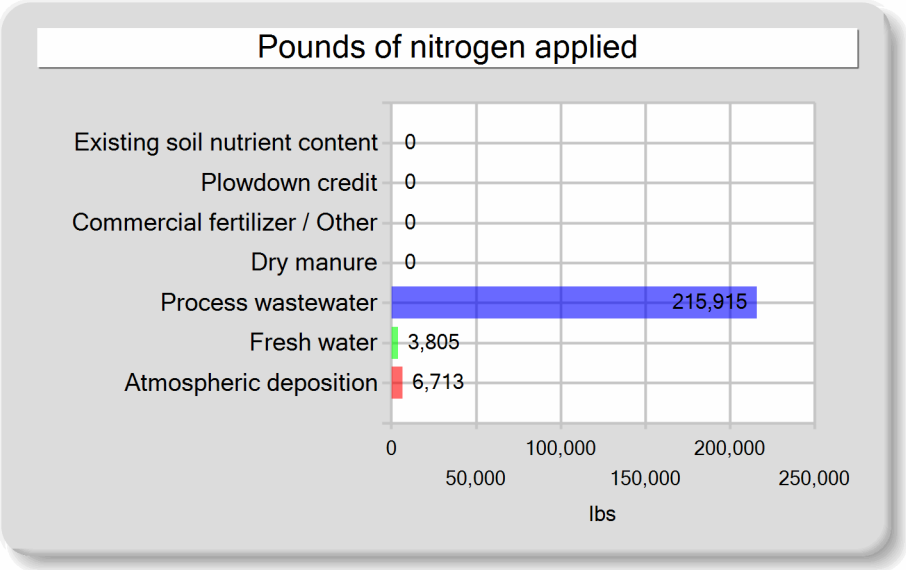
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

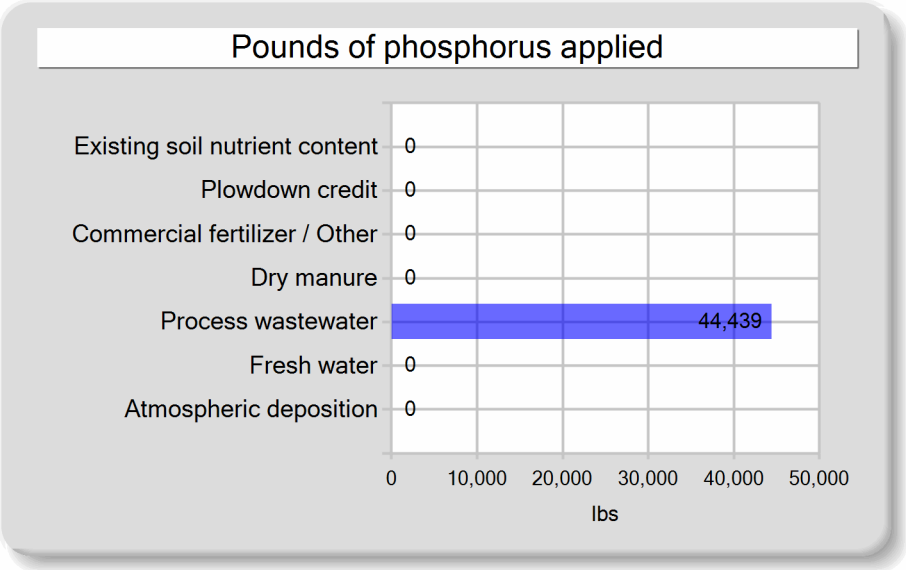
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215,914.58	44,438.63	285,158.66	1,441,671.52
Fresh water	3,804.94	0.00	0.00	273,840.34
Atmospheric deposition	6,713.00	0.00	0.00	0.00
Total nutrients applied	226,432.52	44,438.63	285,158.66	1,715,511.86
Anticipated crop nutrient removal	192,560.00	34,037.00	159,238.00	0.00
Actual crop nutrient removal	181,017.71	21,136.37	130,141.57	954,159.46
Nutrient balance	45,414.81	23,302.26	155,017.09	761,352.40
Applied to removed ratio	1.25	2.10	2.19	1.80

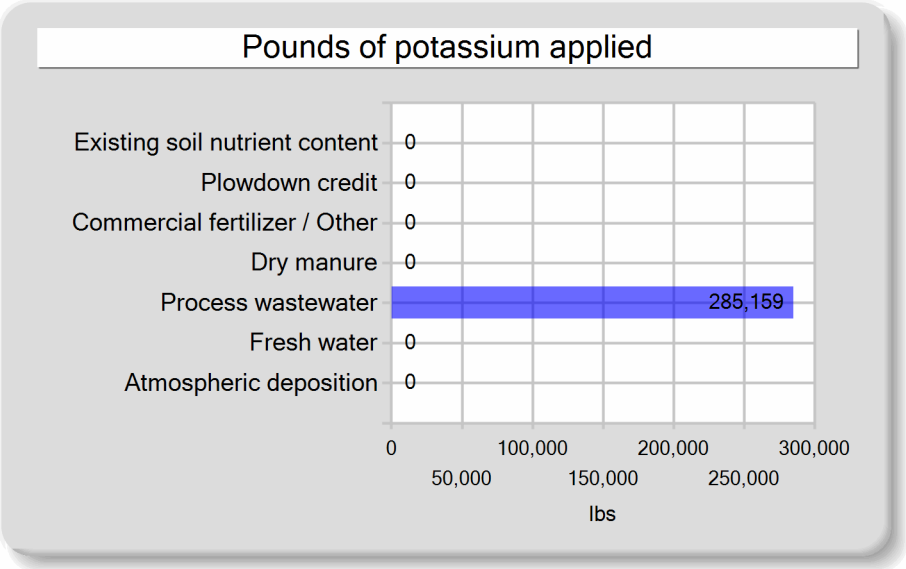
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

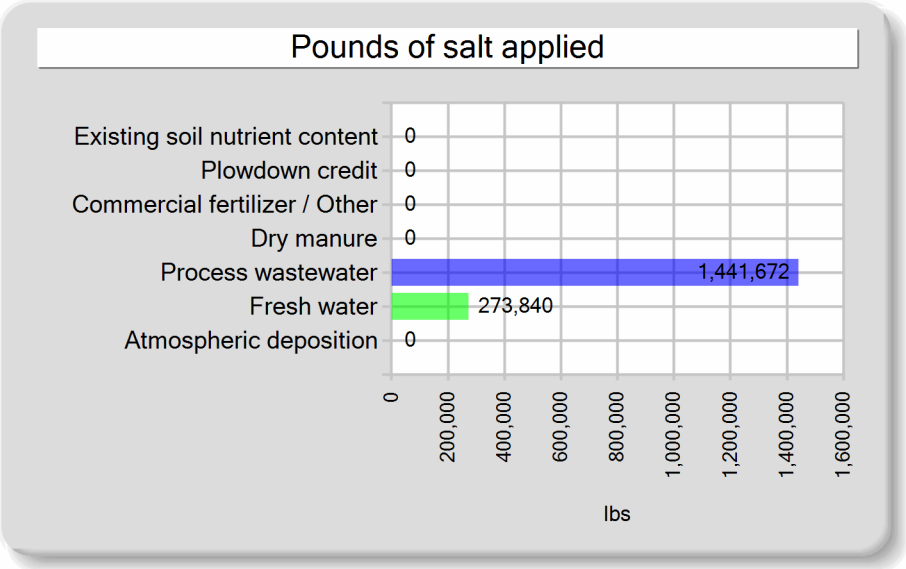


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

X SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
George Andrew teVelde	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
7/12/24	7/12/24
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	See Attachments
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	